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EE9-CS2-1
EE9-FPN2-05

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2017

MSc and EEE/EIE PART IV: MEng and ACGI

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STABILITY AND CONTROL OF NON-LINEAR SYSTEMS

Friday, 12 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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1. Consider the following two-dimensional time-invariant nonlinear system:

$$\begin{aligned} \dot{x}_1 &= -\sin(x_1 - x_2) \\ \dot{x}_2 &= x_2 - x_2^2 + x_1 \end{aligned}$$

a) Discuss existence and uniqueness of solutions. [2]

b) Find the nullclines of the system and sketch them graphically. [3]

c) Identify the regions of the phase-plane in which the nullclines partition $\mathbb{R}^2$ and the orientation of the vector-field in each one of them. [3]

d) Find, if any, those regions that are forward invariant. [3]

e) Find all equilibria of the system. [2]

f) Linearize the system around each equilibrium and discuss the local phase-portrait. [4]

g) Sketch the global phase portrait of the system. [3]

2. Consider the following 3-dimensional, nonlinear system of differential equations:

$$\begin{aligned} \dot{x}_1 &= -\text{atan}(x_1) + x_2 \\ \dot{x}_2 &= -2x_1 + x_3^3 \\ \dot{x}_3 &= -x_3 - x_2 \end{aligned}$$

along with the candidate Lyapunov function given below:

$$V(x) = \frac{x_1^2}{2} + \alpha x_2^2 + \beta x_3^4.$$

a) Prove that for all α and β positive, $V(x)$ is a positive definite and radially unbounded function. [5]

b) Choose α and β positive, so that the function $\dot{V}(x)$ is (at least) negative semi-definite. [5]

c) Show that the origin is a globally asymptotically stable equilibrium. [5]

d) Consider next the system obtained by replacing the equation for $\dot{x}_3$ by the following one:

$$\dot{x}_3 = x_3 + x_2.$$

Show, by using suitable instability criteria, that the origin is unstable.
[*Hint: start out with a Lyapunov function of the following type:* $V(x) = -\frac{x_1^2}{2} + \alpha x_2^2 + \beta x_3^4$ *where* α *and* β *are not constrained in sign. Does* V *meet all assumptions of the instability criterion?*] [5]

3. Consider the following 3-dimensional nonlinear system with state variable $x = [x_1, x_2, x_3]'$ and scalar input d:

$$\begin{aligned} \dot{x}_1 &= \operatorname{atan}(d - x_1) \\ \dot{x}_2 &= -x_2^5 + x_1^2 x_3^2 \\ \dot{x}_3 &= x_1^3 - x_3. \end{aligned}$$

a) Decompose the system as a cascade interconnection of subsystems (this may involve two or more subsystems). [4]

b) Show that each of the subsystems is Input-to-State Stable. [6]

c) Compute a suitable upper-bound of class $\mathscr{K}$ to the nonlinear gain from input d to state x. [6]

d) Show that Input-to-State Stability is a form of dissipativity. What kind of estimate on solutions of an ISS system can be achieved by regarding the corresponding ISS Lyapunov function as a storage function of a dissipative system? [4]

4. Consider the following affine nonlinear control system:

$$\begin{aligned} \dot{x}_1 &= x_1 + x_3 \\ \dot{x}_2 &= -x_2^2 + u \\ \dot{x}_3 &= -x_3 + x_2 + \sin(x_1) \end{aligned}$$

$$y = x_1.$$

a) Compute the relative degree from input u to output y and emphasize whether the notion adopted is local or global. [4]

b) Design an input-output linearizing state feedback. [3]

c) By employing pole placement, find a suitable stabilizing output feedback controller for the linear system resulting from the previous question. [3]

d) Modify the output equation as follows: $y = x_3$. Compute the associated relative degree, from input u to the newly defined output variable and rewrite the system in normal form. [6]

e) Compute the zero-dynamics and comment on their stability. [4]